

E-Commerce Storefront (ROW-32) — MERN Base Code

This PDF contains a lightweight hackathon-ready MERN base code structure for a full-featured e-commerce storefront with: Product catalog, cart, filters, checkout form, order summary, stock display, wishlist, and payment mock.

Project Structure

```
frontend/  
src/  
  components/  
  pages/  
  store/  
  data/  
  backend/  
  models/  
  routes/  
  controllers/  
  server.js
```

Product Model

```
const ProductSchema = new mongoose.Schema({  
  name: String,  
  price: Number,  
  stock: Number,  
  category: String,  
  image: String,  
  rating: Number  
});
```

Cart Store (Zustand)

```
export const useCart = create((set) => ({  
  items: [],  
  addToCart: (product) => set((state) => ({  
    items: [...state.items, product]  
  })),  
}));
```

Product Card

```
addToCart(product)}>  
Add to Cart  
  
Add to Wishlist
```

Filters

```
const filtered = products.filter(  
  p => p.category === selectedCategory  
);
```

Checkout Form

```
UPI  
Card  
COD
```

Order Summary

```
const total = items.reduce(
  (sum, i) => sum + i.price * i.qty,
  0
);
```

Payment Mock

```
setTimeout(() => {
  setStatus('success');
}, 2000);
```

Stock Display

```
{product.stock > 0 ? 'In Stock' : 'Out of Stock'}
```

API Routes

```
GET /api/products
POST /api/orders
GET /api/orders/:id
```

Run Commands

```
npm install
npm run dev
```

Hackathon Checklist

- ✓ Product catalog
- ✓ Search & filters
- ✓ Cart management
- ✓ Wishlist
- ✓ Checkout form
- ✓ Order summary
- ✓ Stock display
- ✓ Payment simulation
- ✓ Responsive UI